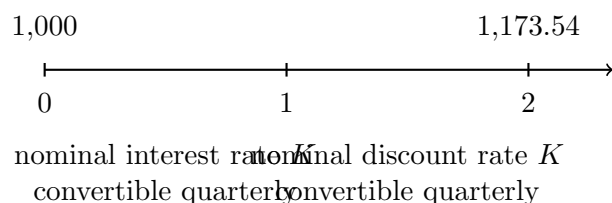


## Problem 11.22

John invests 1,000 in a fund which earns interest during the first year at a nominal rate of  $K$  convertible quarterly. During the second year, the fund earns interest at a nominal discount rate of  $K$  convertible quarterly. At the end of the second year, the fund has accumulated to 1,173.54. Calculate  $K$ .

### Solution

The cash-flow timeline is



During the first year, the quarterly accumulation factor is  $1 + K/4$ . During the second year, the quarterly accumulation factor corresponding to the nominal discount rate is  $(1 - K/4)^{-1}$ . Thus,

$$1,000 \left(1 + \frac{K}{4}\right)^4 \left(1 - \frac{K}{4}\right)^{-4} = 1,173.54,$$

$$\left(\frac{1 + K/4}{1 - K/4}\right)^4 = 1.17354.$$

Taking the fourth root gives

$$\frac{1 + K/4}{1 - K/4} = (1.17354)^{1/4} \approx 1.040817233.$$

Let  $r = (1.17354)^{1/4}$ . Then

$$1 + \frac{K}{4} = r \left(1 - \frac{K}{4}\right),$$

$$\frac{K}{4}(1 + r) = r - 1,$$

$$K = 4 \left(\frac{r - 1}{r + 1}\right)$$

$$\approx 0.08000174.$$

Therefore, to the precision of the accumulated value,

$$\boxed{K \approx 0.08 = 8\%}.$$